

Running the web application in a minikube cluster

Some of the prerequisites for installing minikube on an Ubuntu operating system are:

Containers or virtual machine managers	Architecture
Docker, QEMU, Hyperkit, Hyper-V, KVM, Parallels, Podman, VirtualBox, or VMware Fusion/Workstation	2 CPUs or more 2GB of free memory 20GB of free disk space

To install the latest minikube stable release on Ubuntu 22.04.1 LTS, execute the following commands:

```
osfy@ubuntu:~/Desktop$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube_latest_amd64.deb
% Total    % Received % Xferd Average Speed   Time    Time
Time Current          Dload  Upload   Total   Spent    Left  Speed
100 26.1M  100 26.1M    0     0 1430k    0  0:00:18
0:00:18 --:--:-- 1231k
```

```
osfy@ubuntu:~/Desktop$ sudo dpkg -i minikube_latest_amd64.deb

Selecting previously unselected package minikube.
(Reading database ... 204624 files and directories currently
installed.)
Preparing to unpack minikube_latest_amd64.deb ...
Unpacking minikube (1.28.0-0) ...
Setting up minikube (1.28.0-0) ...
```

Start your minikube cluster using the minikube *start* command. If you face any issues, stop the cluster, delete it and execute the minikube *start* command again. The next step is to verify Java and Maven installation, create a Docker image, and deploy the Docker image using Kubernetes YAML in the minikube cluster.

Next, install Java and Maven in Ubuntu OS. Once installation is completed successfully, verify the Java version using the *java -version* command.

```
osfy@ubuntu:~/Desktop$ java -version

openjdk version "11.0.17" 2022-10-18
OpenJDK Runtime Environment (build 11.0.17+8-post-Ubuntu-
1ubuntu222.04)
OpenJDK 64-Bit Server VM (build 11.0.17+8-post-Ubuntu-
1ubuntu222.04, mixed mode, sharing)
```

Now, verify the Maven version using the *mvn -version* command:

```
osfy@ubuntu:~/Desktop$ mvn -version
```

```
Apache Maven 3.6.3
```

```
Maven home: /usr/share/maven
```

```
Java version: 11.0.17, vendor: Ubuntu, runtime: /usr/lib/jvm/
java-11-openjdk-amd64
```

```
Default locale: en_IN, platform encoding: UTF-8
```

```
OS name: "linux", version: "5.15.0-53-generic", arch:
"amd64", family: "unix"
```

We are going to use Petclinic as a sample application, which is a Spring Boot application built using Maven or Gradle. In this article we will build a JAR file and deploy it in a minikube cluster. Visit <https://github.com/spring-projects/spring-petclinic> to know more. Copy the GitHub repository URL from the above URL, and use the *git clone* command to clone it locally.

```
osfy@ubuntu:~/Desktop$ git clone https://github.com/spring-
projects/spring-petclinic.git
```

```
Cloning into 'spring-petclinic'...
```

```
remote: Enumerating objects: 9424, done.
```

```
remote: Total 9424 (delta 0), reused 0 (delta 0), pack-reused
9424
```

```
Receiving objects: 100% (9424/9424), 7.64 MiB | 1.55 MiB/s,
done.
```

```
Resolving deltas: 100% (3567/3567), done.
```

Once the source code is available, go to the root of the project using the *change directory* command as given below, and execute the *./mvnw package* command to create a JAR file. At first execution it will try to download multiple dependencies, and hence will take some time to complete execution.

```
osfy@ubuntu:~/Desktop$ cd spring-petclinic
osfy@ubuntu:~/Desktop/spring-petclinic$ ./mvnw package
Warning: JAVA_HOME environment variable is not set.
[INFO] Scanning for projects...
Downloading from spring-snapshots:
[INFO]
[INFO] -----< org.springframework.samples:spring-
petclinic >-----
[INFO] Building petclinic 2.7.3
[INFO] -----[ jar ]-----
.
.
[INFO] Building jar: /home/osfy/Desktop/spring-petclinic/
target/spring-petclinic-2.7.3.jar
[INFO]
[INFO] --- spring-boot-maven-plugin:2.7.3:repackage
(repackage) @ spring-petclinic ---
[INFO] Replacing main artifact with repackaged archive
[INFO] -----
```